

similar triangles

a matching activity



instructions

Similar triangles are enlargements of each other.

To be sure that triangles are similar they must have either:

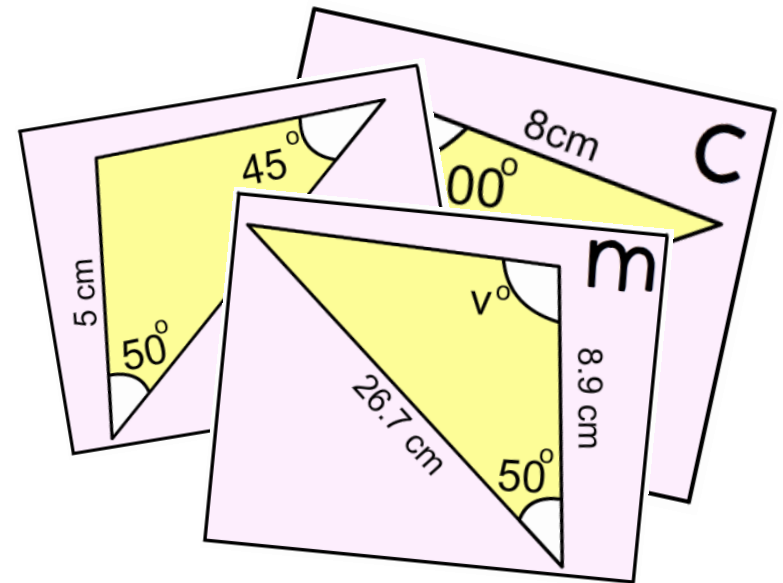
1. AAA—Three Angles the Same
2. SSS—Three sides in the same ratio
3. SAS—An angle in between two sides that are in the same ratio

On the next two pages are a set of 14 triangles.
Match them into pairs that are definitely similar.

Hints:

- (1) To get started, label all the angles in the triangles that are missing, by using the fact that they must add up to 180 degrees.
- (2) Look for triangles that have an angle in common, then check the sides are in the same ratio.

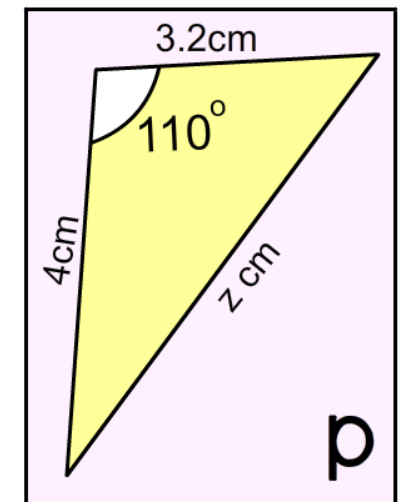
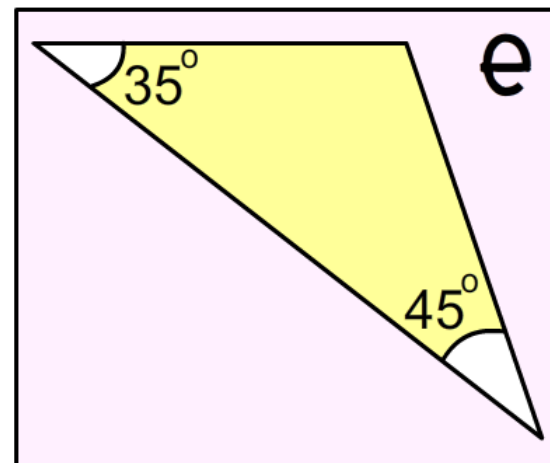
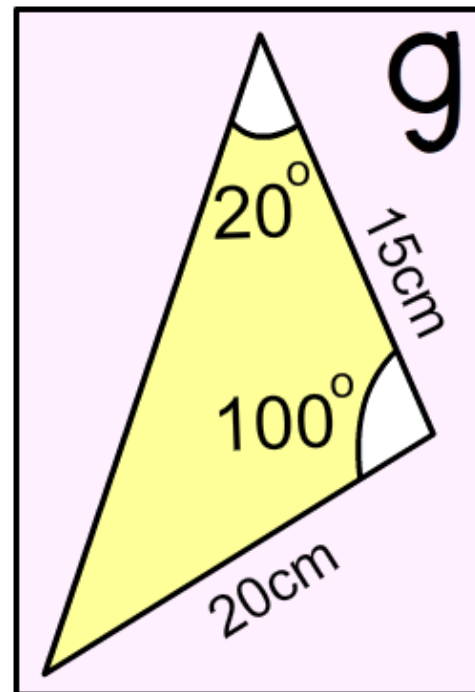
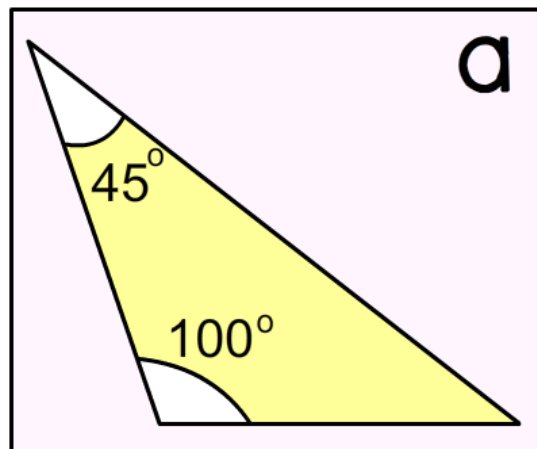
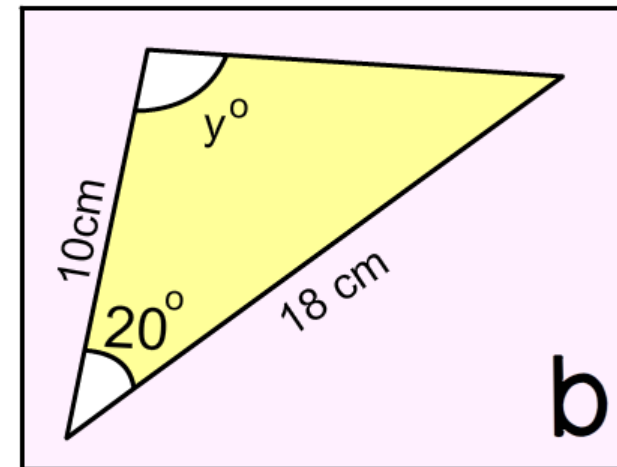
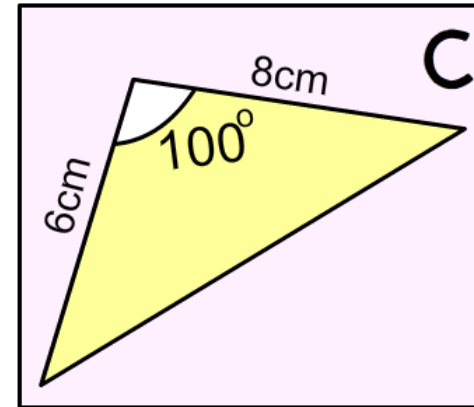
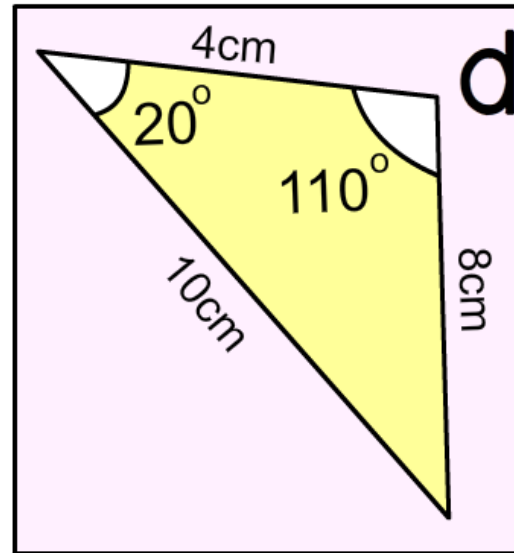
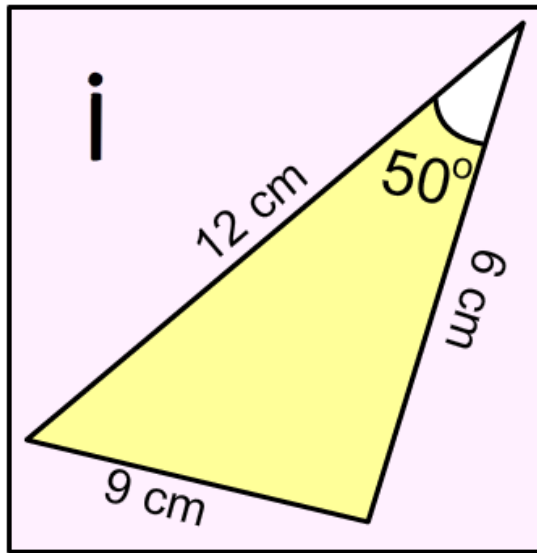
Once you have matched the triangles, record your results in the table. Then work out what the missing angles and sides must be that are labeled v,w,x,y,z.

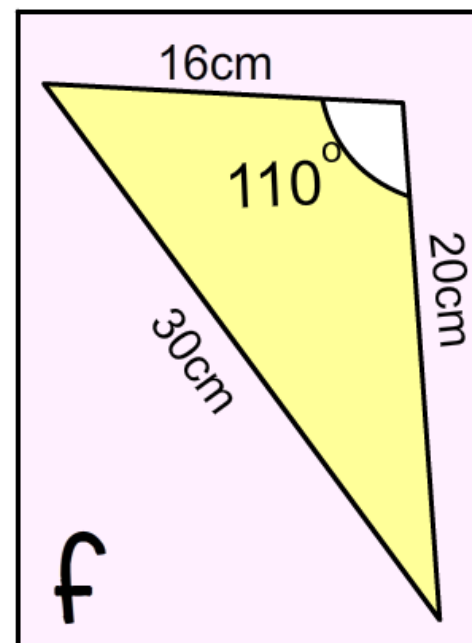
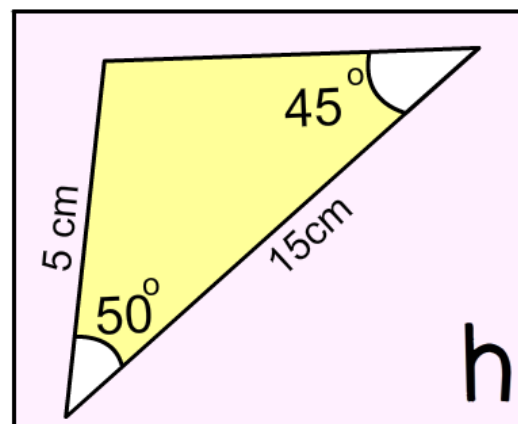
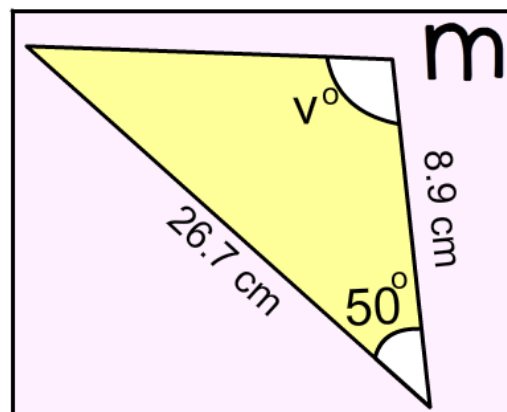
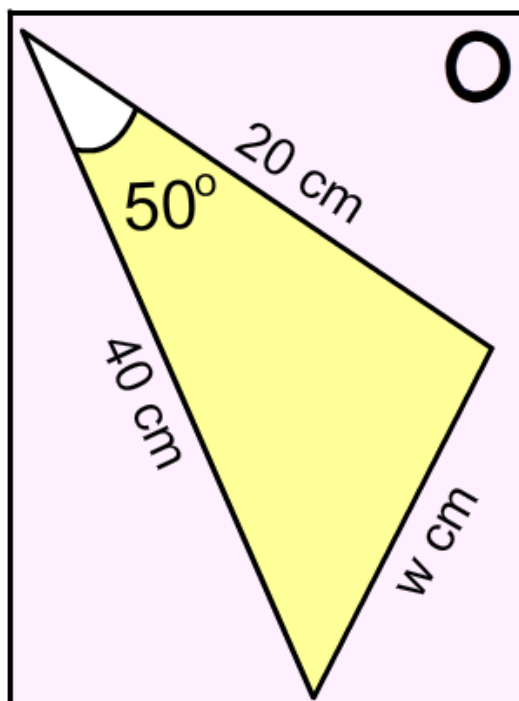
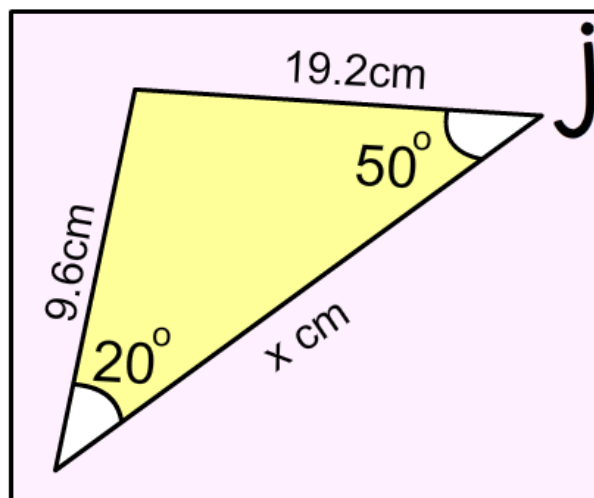
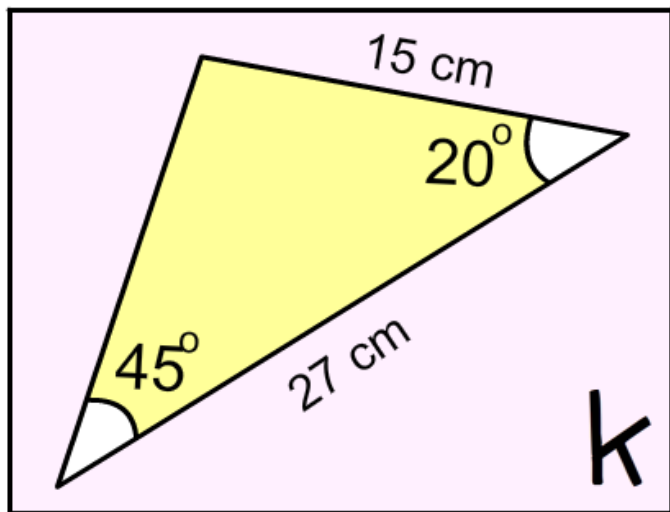


match it up!

these triangles are not drawn accurately.

can you find pairs that are similar?





write your
matching pairs
here

now work out
the missing
sides and
angles:

$v =$ $w =$
 $x =$ $y =$
 $z =$

answers

a	e
b	k
c	g
d	j
f	p
h	m
i	o

now work out the
missing sides and
angles:

$$v = 85 \quad w = 30$$

$$x = 24 \quad y = 115$$

$$z = 6$$